

Tony Menzo

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Academic Appointment

Sep. 2025–present **Postdoctoral Fellow, Particle Theory and Phenomenology**
Joint appointment: Fermi National Accelerator Laboratory and the University of Alabama

Research Interests

Particle phenomenology: rare lepton decays, dark sectors, and effective field theory; hadronization and event generation using machine learning and optimal transport; agentic tools for symbolic computation and high-energy physics workflows.

Education

2020–2025 **Ph.D. in Physics**, University of Cincinnati
Advisor: Jure Zupan
Dissertation: *Rare Lepton Decays and Differentiable Hadronization Models: From Signatures of New Physics to Data-driven Event Generation*

2014–2019 **Bachelor of Arts and Sciences in Physics**, The Ohio State University
Minors: Earth Science (Geophysics) and Philosophy; research mentor: Stuart Raby

Fellowships and Awards

2024–2025 **URA Visiting Scholars Program**, Fermilab
Time-dependent signals at the intensity frontier; one-month visit

2024 **URA Visiting Scholars Program**, Fermilab
Exploring exotic signatures at Mu2e; three-month visit

2023–2024 **DOE Office of Science Graduate Student Research (SCGSR)**
Lawrence Berkeley National Laboratory; six-month research placement

2023 **First place**, University of Cincinnati Physics Poster Competition

2022 **Sandwich Scholarship**, Technion–Israel Institute of Technology
Three-month visiting research studentship

Teaching and Mentoring

Summer 2026 **Mentor**, Google Summer of Code (HEPSIM)

Apr. 2026 **Tutorial instructor**, MuonBridge hands-on tutorial, MIAPbP

Jan. 2026 **Tutorial instructor**, (HEP-focused) Agentic programming, CMSDAS, Fermilab

Jun. 2025 **Tutorial instructor**, CTEQ + MCgen summer workshop and training program

2020–2021 **Teaching Assistant**, Physics Department, University of Cincinnati

Summer 2019 **Physics Instructor**, Upward Bound, Columbus State Community College

2015–2019 **Physics and Mathematics Tutor**, The Ohio State University
Office of Diversity and Inclusion and Physics Department

Publications and Preprints

1. **Tony Menzo**, George T. Fleming, Konstantin T. Matchev, Stephen Mrenna, Alexander Roman. “Reining in an Agentic Harness for High Energy Physics.” Preprint. <https://arxiv.org/abs/2609.00107> (2026).
2. Benoit Assi, Christian Bierlich, Rikab Gambhir, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Vistas: A Visualization Interface for Particle Collision Simulations.” Preprint. <https://arxiv.org/abs/2606.19524> (2026).
3. Yuval Grossman, **Tony Menzo**, Stefan Schacht, Chinh-san Sieng, Jure Zupan. “Unbinned extraction of γ from $B \rightarrow DK$ with normalizing flows.” Preprint. <https://arxiv.org/abs/2605.06768> (2026).
4. Yohei Ema, Patrick J. Fox, Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Anupam Ray, Jure Zupan. “New physics in multi-lepton tau decays.” Preprint. <https://arxiv.org/abs/2604.01465> (2026).
5. **Tony Menzo**, Alexander Roman, George T. Fleming, Sergei Gleyzer, Konstantin T. Matchev, Stephen Mrenna. “Agentic Diagrammatica: Towards Autonomous Symbolic Computation in High Energy Physics.” Preprint. <https://arxiv.org/abs/2603.26990> (2026).
6. **Tony Menzo**. “Conservation laws and effective hadronization models.” Preprint. <https://arxiv.org/abs/2602.12599> (2026).
7. Benoît Assi, Christian Bierlich, Rikab Gambhir, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Jure Zupan. “HDSense: An efficient method for ranking observable sensitivity.” Preprint. <https://arxiv.org/abs/2602.01509> (2026).
8. **Tony Menzo**, Alexander Roman, Sergei Gleyzer, Konstantin Matchev, George T. Fleming, Stefan H  che, Stephen Mrenna, Prasanth Shyamsundar. “HEPTAPOD: Orchestrating High Energy Physics Workflows Towards Autonomous Agency.” Preprint. <https://arxiv.org/abs/2512.15867> (2025).
9. Anja Butter, Ayodele Ore, Sofia Palacios Schweitzer, Tilman Plehn, Beno  t Assi, Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Iterative HOMER with uncertainties.” SciPost Phys. 20, 042 (2026). <https://arxiv.org/abs/2509.03592> (2025).
10. Yohei Ema, Patrick J. Fox, Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Anupam Ray, Jure Zupan. “Long-lived Axion-Like Particles from Tau Decays.” Phys. Rev. D 112, 115028 (2025). <https://arxiv.org/abs/2507.15271> (2025).
11. Beno  t Assi, Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Post-hoc reweighting of hadron production in the Lund string model.” SciPost Phys. 19, 104 (2025). <https://arxiv.org/abs/2505.00142> (2025).
12. Innes Bigaran, Patrick J. Fox, Yann Gouttenoire, Roni Harnik, Gordan Krnjaic, **Tony Menzo**, Jure Zupan. “Direct Detection of Ultralight Dark Matter via Charged Lepton Flavor Violation.” Phys. Rev. Lett. 136, 251801 (2026). <https://arxiv.org/abs/2503.07722> (2025).

Publications and Preprints (continued)

13. Benoit Assi, Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Characterizing the hadronization of parton showers using the HOMER method.” *SciPost Phys.* 19, 125 (2025). <https://arxiv.org/abs/2503.05667> (2025).
14. Chaja Baruch, Patrick J. Fitzpatrick, **Tony Menzo**, Yotam Soreq, Sokratis Trifinopoulos, Jure Zupan. “Searching for exotic scalars at fusion reactors.” *JHEP* 10 (2025) 215. <https://arxiv.org/abs/2502.12314> (2025).
15. Nick Heller, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Benjamin Nachman, Andrzej Siodmok, Manuel Szewc, Ahmed Youssef. “Rejection Sampling with Autodifferentiation - Case study: Fitting a Hadronization Model.” Preprint. <https://arxiv.org/abs/2411.02194> (2024).
16. Christian Bierlich, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Describing Hadronization via Histories and Observables for Monte-Carlo Event Reweighting.” *SciPost Phys.* 18, 054 (2025). <https://arxiv.org/abs/2410.06342> (2024).
17. Patrick J. Fox, Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Jure Zupan. “Muon-induced baryon number violation.” *Phys. Rev. D* 110, 075015 (2024). <https://arxiv.org/abs/2407.03450> (2024).
18. Wick Haxton, Kenneth McElvain, **Tony Menzo**, Evan Rule, Jure Zupan. “Effective theory tower for $\mu \rightarrow e$ conversion.” *JHEP* 11 (2024) 076. <https://arxiv.org/abs/2406.13818> (2024).
19. Christian Bierlich, Phil Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Towards a data-driven model of hadronization using normalizing flows.” *SciPost Phys.* 17, 045 (2024). <https://arxiv.org/abs/2311.09296> (2023).
20. Christian Bierlich, Philip Ilten, **Tony Menzo**, Stephen Mrenna, Manuel Szewc, Michael K. Wilkinson, Ahmed Youssef, Jure Zupan. “Reweighting Monte Carlo Predictions and Automated Fragmentation Variations in Pythia 8.” *SciPost Phys.* 16, 134 (2024). <https://arxiv.org/abs/2308.13459> (2023).
21. Matheus Hostert, **Tony Menzo**, Maxim Pospelov, Jure Zupan. “New physics in multi-electron muon decays.” *JHEP* 10 (2023) 006. <https://arxiv.org/abs/2306.15631> (2023).
22. Adam Davis, **Tony Menzo**, Ahmed Youssef, Jure Zupan. “Earth mover’s distance as a measure of CP violation.” *JHEP* 06 (2023) 098. <https://arxiv.org/abs/2301.13211> (2023).
23. Reuven Balkin, Eric Madge, **Tony Menzo**, Gilad Perez, Yotam Soreq, Jure Zupan. “On the implications of positive W mass shift.” *JHEP* 05 (2022) 133. <https://arxiv.org/abs/2204.05992> (2022).
24. Phil Ilten, **Tony Menzo**, Ahmed Youssef, Jure Zupan. “Modeling hadronization using machine learning.” *SciPost Phys.* 14, 027 (2023). <https://arxiv.org/abs/2203.04983> (2022).

Research Software

AGENT FRAMEWORKS AND TOOLKITS

HEPTAPOD — *Maintainer*. Agentic workflows for symbolic calculations, event generation, and HEP analysis. Related publications: [5, 8].

<https://github.com/tonymenzo/heptapod>

Orchestral — *Co-maintainer*. A framework and provider-agnostic harness for LLM agent orchestration across model providers. Related paper: <https://arxiv.org/abs/2601.02577>.

<https://orchestral-ai.com>

TOOLBASE — *Maintainer*. Packaging, distribution, and deployment of tools and skills for AI agents.

<https://github.com/alex314/toolbase>

TOOLBENCH — *Maintainer*. A framework for benchmarking agentic tools and execution environments.

<https://github.com/tonymenzo/toolbench>

PARTICLE PHYSICS AND COMPUTATIONAL METHODS

MLHAD — Machine learning methods for hadronization modeling and reweighting. Related publications: [9, 11, 13, 15, 16, 19, 24].

<https://gitlab.com/uchep/mlhad>

RSA — Rejection sampling with automatic differentiation. Related publication: [15].

<https://github.com/tonymenzo/RSA>

EMD4CPV — Optimal transport methods for detecting CP violation. Related publication: [22].

<https://github.com/adamdddave/emd4cpv>

MuonBridge — Effective field theory matching and predictions for $\mu \rightarrow e$ conversion. Related publication: [18].

<https://github.com/Berkeley-Electroweak-Physics/MuonBridge>

EDUCATIONAL

MCgen tutorials — Contributions to Monte Carlo event-generation tutorials.

<https://gitlab.com/mcgen-ct/tutorials>

Academic Service

Referee, Journal of High Energy Physics (JHEP); PRX Intelligence.

Co-organizer, Machine Learning Journal Club, University of Cincinnati, 2024–present.

Organizing committee member, Agentic AI for BSM Event Generation and Analysis Workshop at the LPC, Fermilab, September 9–11, 2026.

<https://indico.fnal.gov/event/75639/>

Co-organizer, University of Alabama HEP Seminar, 2025–present.

Seminars and Colloquia

Slides: <https://tonymenzo.com/docs/presentations>

Sep. 2026	Portable agentic phenomenology <i>University of Cincinnati HEP/Astro Seminar</i>
Sep. 2026	Portable agentic phenomenology <i>Colorado State University HEPPA Seminar</i> (invited)
Apr. 2026	Agentic phenomenology <i>BNL HEP Theory Seminar</i> (invited)
Mar. 2026	Conservation laws and effective (string-)hadronization models <i>Monash HEP Seminar</i> (invited)
Jan. 2026	Agents in HEP(-ex) <i>Fermilab Neutrino Division</i>
Jan. 2026	Agents in HEP(-ph/-th) <i>Fermilab Theory Division</i> (chalk talk)
Oct. 2025	The journey towards differentiable hadronization models <i>University of Alabama HEP Seminar</i>
Sep. 2025	The journey towards differentiable hadronization models <i>Michigan State University HEP Seminar</i> (invited)
Jul. 2025	Differentiable hadronization models, heavy and light new physics in rare lepton decays, and more... <i>University of Alabama, Tuscaloosa</i>
Nov. 2024	Rare μ decays: high-intensity probes of heavy and light new physics <i>University of Maryland EPT Seminar</i>
Nov. 2024	Data-driven hadronization models <i>Rutgers NHETC Seminar</i>
Nov. 2024	From heavy to light: new physics @ high-intensity μ experiments <i>Cornell University LEPP Theory Seminar</i>
Oct. 2024	From heavy to light: new physics @ high-intensity μ experiments <i>SLAC Theory Seminar</i>
Oct. 2024	From heavy to light: new physics @ muon factories <i>UCSD HEP Seminar</i>
Oct. 2024	From heavy to light: new physics @ muon factories <i>UCSB HEX-HET Seminar</i>
May 2024	Building bridges: an effective field theory for muon-to-electron conversion <i>Berkeley Nuclear Theory Seminar</i>
Feb. 2024	Optimal transport and CP violation: the Dalitz Mover's Distance <i>Berkeley 4D Seminar, University of California, Berkeley</i>
Oct. 2023	Micro from macro (and applications in event generators) <i>ATLAS Theory Lunch, Lawrence Berkeley National Laboratory</i>
Sep. 2023	Towards a data-driven model of hadronization <i>IAIFI Journal Club, MIT</i>
May 2022	Modeling hadronization using machine learning <i>Technion HEP Journal Club</i>

Conference and Workshop Presentations

Slides: <https://tonymenzo.com/docs/presentations>

Sep. 2026	Harness HEPineering <i>Agentic AI for BSM Event Generation and Analysis Workshop, Fermilab LPC</i>
Sep. 2026	Benchmarking with HEPTAPOD <i>Agentic AI for BSM Event Generation and Analysis Workshop, Fermilab LPC</i>
Jun. 2026	Detecting Ultralight DM via CLFV <i>CLFV 2026</i> (invited)
Jun. 2026	Agentic systems for hep-ph <i>AI Reasoning in Theoretical Physics, Aspen (chalk talk)</i>
May 2026	Toward agentic phenomenology <i>PHENO 2026</i>
Jan. 2026	AmSC briefing <i>Online</i>
Dec. 2025	Agents in HEP <i>University of Miami Physics Conference</i>
Jun. 2025	A flavorful and dark cascade for low-scale leptogenesis <i>NPN 2025</i>
Jun. 2025	Direct detection of ultralight dark matter with charged-lepton-flavor-violation <i>FPCP 2025</i>
May 2025	Road to "differentiable" Pythia? + Half-baked thoughts on energy conservation in hadronization <i>Pythia Week @ Lund University</i>
May 2025	Direct detection of ultralight dark matter with charged-lepton-flavor-violation <i>LULBI @ Weizmann Institute</i>
Apr. 2025	Direct detection of ultralight dark matter with charged-lepton-flavor-violation <i>PIKIMO</i>
Sep. 2024	Signatures of light new physics @ muon factories <i>NuFact 2024</i>
May 2024	UV imprints on muon-to-electron conversion <i>PHENO 2024</i>
May 2024	Multi-electron muon decays ($\mu \rightarrow eeeee$) <i>PIKIMO</i>
May 2023	Earth Mover's Distance and CP-violation <i>PHENO 2023</i>
Apr. 2023	Dark low-scale leptogenesis <i>PIKIMO, The Ohio State University</i>
Mar. 2023	Modeling hadronization using machine learning <i>DIS 2023</i>
Nov. 2022	Modeling hadronization using machine learning <i>QCD@LHC 2022</i>
Nov. 2022	The Wasserstein distance for CP-violation <i>PIKIMO 13</i>
Oct. 2022	Modeling hadronization using machine learning <i>AI4EIC</i>
Jul. 2022	Modeling hadronization using machine learning <i>RBSM Summer School</i>